
Modular Programming – Practical Work

Week 1

Recap exercises from 631-1

Visited concepts: loops, arrays, code understanding

Notations: W2Q1 = Week 2, Questions of type 1

Question types:

- Type 1: theory, concepts *knowledge*
- Type 2: understanding typically based on code reading *understanding*
- Type 3: producing yourself code *know-how*

A moodle deposit space is open every week for you to upload your solutions. Solutions of the exercises are provided one week later.

Important points:

- 602_F and 602_D classes should upload solutions **before Wednesday 0h00 the following week**
- **2 or 3 practical work will be corrected** and will count for one third of the intermediate test note

Conventions about fonts:

- `Lucida Console` is used for Java reserved commands
- `Courier` is used for the code
- Arial is used for the text

W1Q1 – 1

1. What is the difference between an algorithm and a program?

A program is the **translation** of an algorithm in a language that can be understood by the computer.

2. Why does the expression $5+3*2$ gives 11 and not 16? How does a programming language handle this?

In all programming language including Java, there is a **precedence** of the mathematical operators when the computer evaluates an expression. In the case of the example, the multiplication $*$ has more priority than the addition $+$.

W1Q1 – 2

1. What is happening if you define a for loop like this:

```
for(short i=0;i<100000;i++)
```

Variable i is of **type short**, which means that its range of potential values is in between -32 768 and 32 767. When the increment of variable i reaches the maximal value, it simply starts again from the minimal value, i.e. -32 768. The consequence is that the stop condition $0<100000$ is never met and the loop becomes infinite.

2. I want to define a 2 dimensional int array as follows:

- 3 rows
- row 1 has 6 columns, row 2 has 5 columns and row 3 has 4 columns

Is it possible in Java?

Yes, it is possible to define multidimensional arrays with different number of columns in each row:

```
int [][]x = new int[3][];  
x[0] = new int[6];  
x[1] = new int[5];  
x[2] = new int[4];
```

W1Q2 – 1

What does the following code display? Underline the correct answer.

```
String myString ;  
myString = "jean-claude" ;  
myString.replace('j', 'J') ;  
System.out.println(myString) ;
```

a) jean-claude

b) Jean-claude

W1Q2 – 2

What does this program display? Give the answer without running this code. Read the code as if you were Eclipse's debugger.

```
public class X {  
  
    public static void main(String[] args) {  
        int array[] = {8,3,5,6,4,7} ;  
        mystery(array) ;  
        display(array) ;  
    }  
  
    public static void mystery(int array[])  
    {  
        int length=array.length;  
        boolean permut;  
  
        do  
        {  
            permut=false;  
            for(int i=0;i<length-1;i++)  
            {  
                if(array[i]>array[i+1])  
                {  
                    move(array,i,i+1);  
                    permut=true;  
                }  
            }  
        }  
        while(permut);  
    }  
  
    public static void move(int array[], int pos1, int pos2)  
    {  
        int temp = array[pos1] ;
```

```

        array[pos1] = array[pos2] ;
        array[pos2] = temp ;
    }

    public static void display(int [] array)
    {
        for (int i = 0 ; i<array.length ; i++)
            System.out.print(array[i] + " ");
    }
}

```

This program sorts an array of int. (Bubble sort : http://en.wikipedia.org/wiki/Bubble_sort)

The result is: **3 4 5 6 7 8**

W1Q2 – 3

What does this method display if we have “hello” and 3 for input? Please give the answer without running this code or checking the results on the internet. Read the code as if you were Eclipse’s debugger.

```

public static String mystery(String input, int shiftValue)
{
    String result = "";

    char[] charArray = input.toCharArray();
    int[] outputArray = new int[charArray.length];

    for( int i=0; i<charArray.length; i++ )
    {
        if(charArray[i] == ' ')
        {
            outputArray[i] = charArray[i];
        }
        else
        {
            int shifted = (charArray[i] + shiftValue);
            outputArray[i] = shifted ;
        }
        result += Character.toString((char) outputArray[i]);
    }
    return result;
}

```

This is an encryption method (Caesar Algorithm : http://en.wikipedia.org/wiki/Caesar_cipher)

This program displays: **Khoor**

You can now start Eclipse. Q3 means questions of type 3 that involve programming with Eclipse.

W1Q3 – 1

Read and understand what the following code does and then implement the same result using while loops instead of for loops.

```
for (int i = 1; i < 3; i++ )
{
    for (int j = 3; j > 0; j-- )
    {
        for ( int k = 1; k <= 6; k=k+2 )
            System.out.print( '*' );

        System.out.println();
    }
    System.out.println();
}
```

That program displays :

```
***
***
***

***
***
***
```

Using while loops the result is :

```
int i, j, k ;
i = 1 ;
while (i < 3)
{
    j = 3 ;
    while (j > 0 )
    {
        k = 1 ;
        while (k <= 6 )
        {
            System.out.print( '*' );
            k = k+2 ;
        }
        System.out.println();
        j-- ;
    }
    System.out.println();
    i++ ;
}
```

W1q31.java

W1Q3 – 2

Write an application that plays "guess my number" as follows: Your application chooses the number to be guessed by selecting randomly an integer in the range 1 to 1000. The application then displays the following text:

```
I have a number between 1 and 1000. Can you guess my number?  
Please enter your first guess.
```

The user is entering an integer using the keyboard and the program displays either "Too High" or "Too Low" according to the value of the guess. When the user gets the correct answer, "Correct!" should be displayed.

W1q32.java

W1Q3 – 3

Go back to the W1Q2-2 question and write the decrypt method that will permit you to have the first input back. In that case, it will permit you to have the word "hello" again.

CesarCipher.java

W1Q3 – 4

Table.java
Restaurant.java